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Estimating pharmacokinetic parameters using deep learning

Abstract

A method and system for predicting at least one pharmacokinetic parameter of an agent administered to a subject. One or more processors train, by one or more processors, a neural network based on a simulated training data collection. The simulated training data collection comprising a simulated time-series concentration dataset and a simulated value for a pharmacokinetic parameter that corresponds to the simulated time-series concentration dataset. The one or more processors receive a time-series concentration dataset of the agent obtained from a subject. The one or more processors predict a value for the pharmacokinetic parameter using the time-series concentration dataset and the neural network that has been trained.

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Background/Summary

(1) The present application is a continuation of International Application No. PCT/US2021/023788, filed Mar. 23, 2021, and entitled, “Estimating Pharmacokinetic Parameters Using Deep Learning,” which claims priority to U.S. Provisional Application No. 62/993,639, filed Mar. 23, 2020, and entitled “Estimating Pharmacokinetic Parameters,” the entirety of each of which is incorporated by reference herein.

FIELD
(1) This description is generally directed towards systems and methods for estimating or predicting pharmacokinetic properties for therapeutics. More specifically, there is a need for machine learning-based systems and methods for accurately estimating or predicting pharmacokinetic properties for therapeutics that are administered to patients are disclosed herein.
BACKGROUND
(2) The development of new drugs (e.g., therapeutics) is driven by progress in many disciplines. Such disciplines include drug discovery, biotechnology, and in vivo and in vitro pharmacological/toxicological characterization techniques. Before a new therapeutic can move from a molecule or protein in the laboratory to become a new product in the hospital/clinic or local pharmacy, various questions must be answered with respect to the efficacy, administration, safety, and side effects associated with the therapeutic. Answering these types of questions typically involves a series of clinical trials, which are carefully designed to study the various facets of a new drug candidate.
(3) Pharmacokinetics (PK) and pharmacodynamics (PD) are scientific disciplines associated with therapeutic development that typically involve mathematical modeling. In popular terms, PK (or pK) is often described as “what the body does to the drug” and PD (or pD) as “what the drug does to the body.” More specifically, PK may focus on modeling how the body acts on the drug once it is administered and is subjected to the four bodily processes of absorption, distribution, metabolism and elimination or excretion (ADME). This may be accomplished by modeling concentrations in the body generally or in various areas of the body as a function of time. PD aims at linking these modeled drug concentrations to certain drug effects through a PD-model specifically designed to evaluate those effects. PK/PD modeling may thus link systemic drug concentration kinetics to the resulting drug effects over time. Such modeling enables the description and prediction of the time course of various physiological effects (e.g., tumor cell count, platelet count, neutrophil count, etc.) in response to various dosage regimens.
(4) Conventional mathematical modeling methodologies for PK/PD evaluation may require

iterations of model evaluation and refinement, with human judgement involved in various steps within the loop. This can be time and labor intensive. Examples of such existing mathematical algorithms include expectation-maximization, genetic algorithms, and scatter search. These techniques may be optimization-based, which in practice may mean that the scientist creating the model performs many function and gradient evaluations involving significant trial-and-error. Accordingly, effectively using these existing mathematical techniques to model PK and PD involves a significant amount of know-how and computational time. The know-how prerequisite and computational resource requirement represent significant obstacles along the path towards the broad adoption of PK, PD, and PK/PD modeling for non-expert users.

SUMMARY

(5) In one or more embodiments, a method is provided for predicting at least one pharmacokinetic parameter of an agent administered to a subject. One or more processors train, by one or more processors, a neural network based on a simulated training data collection. The simulated training data collection comprising a simulated time-series concentration dataset and a simulated value for a pharmacokinetic parameter that corresponds to the simulated time-series concentration dataset. The one or more processors receive a time-series concentration dataset of the agent obtained from a subject. The one or more processors predict a value for the pharmacokinetic parameter using the time-series concentration dataset and the neural network that has been trained.

(6) In one or more embodiments, a system is provided for predicting at least one pharmacokinetic parameter of an agent administered to a subject. The system comprises a data store for storing a time-series concentration dataset of the agent obtained from a subject; a computing device communicatively connected to the data store, and a display system communicatively connected to the computing device. The computing device comprises a pharmacokinetic data simulation engine and a pharmacokinetic prediction engine. The pharmacokinetic data simulation engine is configured to generate a simulated training data collection of one or more agents. The simulated training data collection comprises a simulated time-series concentration dataset and a simulated value for a pharmacokinetic parameter that corresponds to the simulated time-series concentration dataset. The pharmacokinetic prediction engine is configured to train a neural network using the simulated training data collection and predict a value for the pharmacokinetic parameter based on the time-series concentration dataset using the neural network that has been trained. The display system is configured to display a report containing the value predicted for the pharmacokinetic parameter.

(7) In one or more embodiments, a method is provided for training a neural network to predict at least one pharmacokinetic parameter of an agent. One or more processors receive a simulation parameter for a compartmental-based pharmacokinetic model. The one or more processors generate a simulated training data collection using the compartmental-based pharmacokinetic model. The simulated training data collection comprises a simulated time-series concentration dataset of one or more agents and a simulated value for a pharmacokinetic parameter that corresponds to the simulated time-series concentration dataset. The compartmental-based pharmacokinetic model is encoded with the received simulation parameters. The one or more processors then train a neural network with the simulated training data collection.

(8) In one or more embodiments, a system for training a neural network to predict a pharmacokinetic parameter value of an agent is disclosed. The system includes a data store, a computing device, and a display system. The data store stores simulation parameters for a compartmental-based pharmacokinetic model. The computing device is communicatively connected to the data store and hosts a pharmacokinetic data simulation engine and a pharmacokinetic prediction engine. The pharmacokinetic data simulation engine is configured to use the compartmental-based pharmacokinetic models to generate a simulated training data collection. The simulated training data collection comprises a simulated time-series concentration dataset of one or more agents and a simulated value for a pharmacokinetic parameter that

corresponds to the simulated time-series concentration dataset. The pharmacokinetic prediction engine is configured to train a neural network using the simulated training data collection.
(9) In one or more embodiments, a method is provided for predicting at least one pharmacokinetic parameter of an agent administered to a subject. One or more processors receive a time-series concentration dataset of the agent obtained from a subject. The one or more processors predict a value for a pharmacokinetic parameter based on the time-series concentration dataset and a neural network that has been trained using a simulated training data collection.
(10) In one or more embodiments, a system is provided for predicting at least one pharmacokinetic parameter of an agent administered to a subject is disclosed. The system includes a data store for storing a time-series concentration dataset of the agent obtained from a subject, a computing device communicatively connected to the data store, and a display system communicatively connected to the computing device. The computing device is configured to train a neural network using a simulated training data collection and predict a value for a pharmacokinetic parameter based on the time-series concentration dataset and the neural network that has been trained. The display system is configured to display a report containing the value predicted for the pharmacokinetic parameter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) For a more complete understanding of the principles disclosed herein, and the advantages thereof, reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:
(2) FIG. **1** is a visualization graph that depicts a standard time-series concentration plot **100** and illustrates examples of exemplary PK parameters.
(3) FIG. **2** is an illustration of a time-series concentration plot.
(4) FIG. **3** is an illustration of a workflow for the sum-of-trapezoids calculation performed for the time-series concentration plot in FIG. **2**.
(5) FIG. **4** is a block diagram of a pharmacokinetic (PK) evaluation system **400** in accordance with one or more example embodiments.
(6) FIG. **5** is a depiction of an exemplary NN system and how it can be used to process time-series concentration data to predict PK parameters, in accordance with various embodiments.
(7) FIG. **6** illustrates an exemplary high-level workflow of how a NN model can be trained during a learning phase and then puts what it has learned into practice to predict PK parameters, in accordance with various embodiments.
(8) FIG. **7** is an illustration of a plot showing percent validation error in accordance with an example embodiment.
(9) FIG. **7** is a schematic diagram of a system for predicting a PK parameter value of an agent (e.g., drug, biologic, etc.) administered to a subject (i.e., patient), in accordance with various embodiments.
(10) FIG. **8** is an illustration of a plot showing an example of training and validation loss as a function of the number of training iterations in accordance with an example embodiment.
(11) FIG. **9** is an illustration of an example training workflow in accordance with various embodiments.
(12) FIG. **10** is an illustration of a series of plots for various forms of compartment modeling in accordance with various embodiments.
(13) FIG. **11** illustrates a set of plots that demonstrates the improved accuracy of estimating AUC and C.sub.max using the neural network approach described herein as compared to the conventional Trapezoidal Rule in accordance with an example embodiment.
(14) FIG. **10** is an exemplary flowchart showing a method for predicting a PK parameter value of

an agent (e.g., drug, biologic, etc.) administered to a subject (i.e., patient), in accordance with various embodiments.
(15) FIG. **11** are plots that demonstrates the improved accuracy of estimating AUC and C.sub.max using a novel NN approach versus the conventional Trapezoidal Rule, in accordance with various embodiments.
(16) FIG. **12** is an exemplary flowchart showing a method for predicting a PK parameter value of an agent (i.e., drug or biologic) administered to a subject (i.e., patient), in accordance with various embodiments.
(17) FIG. **13** is an exemplary flowchart showing a method for training a neural network to predict a pharmacokinetic parameter value of an agent (i.e., drug or biologic), in accordance with various embodiments.
(18) FIG. **14** is an exemplary flowchart showing a method for predicting a PK parameter value of an agent (e.g., drug, biologic, etc.) administered to a subject (i.e., patient), in accordance with various embodiments.
(19) FIG. **15** is a block diagram that illustrates a computer system, upon which embodiments of the present teachings may be implemented.
(20) It is to be understood that the figures are not necessarily drawn to scale, nor are the objects in the figures necessarily drawn to scale in relationship to one another. The figures are depictions that are intended to bring clarity and understanding to various embodiments of apparatuses, systems, and methods disclosed herein. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts. Moreover, it should be appreciated that the drawings are not intended to limit the scope of the present teachings in any way.

DETAILED DESCRIPTION

I. Overview

(21) The principles of pharmacokinetics/pharmacodynamics (PK/PD) have become a well-established quantitative framework for understanding the dose-concentration-effect relationships of various therapeutics and selecting the proper protocols (e.g., dosage, schedules, etc.) for such therapeutics. Current methods for estimating pharmacokinetics properties (e.g., as described below in Section III) are rudimentary. For example, some currently available methods use a non-compartmental analysis (NCA) approach that numerically infers pharmacokinetic parameters. One such method estimates the area under the curve (AUC) formed by the amount of a therapeutic in a subject's blood over time. The AUC can be computed using the Trapezoidal Rule in which the AUC is computed as a sum of trapezoids. These types of approaches are simple and fast.
(22) However, such currently available methodologies may be inaccurate in various situations as their accuracy relies on crucial assumptions or presumptions. These presumptions include, for example, that all the relevant measurements are available and that the time intervals between the measurements are appropriately short and condensed. Therefore, approximation of the AUC using the Trapezoidal Rule can lead to limitations with respect to the accuracy with which pharmacokinetic parameters are estimated. Further, estimation variability of the linear trapezoidal rule strongly depends on data completeness. If missing interim values are imputed by means of linear interpolation, large amounts of missing values will lead to a decreasing estimation accuracy.
(23) Thus, recognizing and taking into account the above-described issues, the specification describes various exemplary embodiments of systems and methods for accurately estimating or predicting pharmacokinetic properties for therapeutics that are administered to patients. It should be appreciated, however, that although the systems and methods disclosed herein refer to their application in pharmacokinetics specifically, they are equally applicable to other analogous fields such as toxicokinetics and toxicodynamics.
(24) The embodiments described herein provide methods and systems for predicting pharmacokinetic parameters with a higher level of accuracy when compared to conventional methods. Using machine learning (ML) or deep learning (DL) helps enable this improved accuracy

with pharmacokinetic modeling. For example, with a deep learning system (e.g., comprising one or more neural network models), predictions of pharmacokinetic parameters are driven by learning features of the underlying data. Further, a deep learning system enables exploitation of local patterns in data to make inferences. Further, with a deep learning system, as the amount of data used for training the deep learning system increases, the prediction accuracy of the deep learning system increases.
(25) In various embodiments, a method is provided for applying one or more mathematical models to generate simulated data; training one or more machine learning (ML) or deep learning (DL) algorithms (e.g., convolutional neural network, etc.) with the simulated PK data for one or more existing or potential future agents (e.g., chemical therapeutic or toxin, biologic therapeutic or toxin, etc.); and generating predictions or providing estimates of one or more PK parameters (e.g., AUC, C.sub.max, T.sub.max, t.sub.1/2, etc.) using real-life (e.g., clinical) PK data.
(26) In various embodiments, a system for predicting or estimating PK parameters can include a data store for storing measured (or "real-life"/clinical) PK data and one or more computers/servers that can host and execute software code that comprises a PK data simulation engine and a PK prediction engine. The PK data simulation engine can be configured to apply one or more mathematical models to simulate PK data that can be used to train ML or DL algorithms. The PK prediction engine can be configured to train the ML or DL algorithms, using the simulated PK data, to make predictions or estimates of PK parameters using the measured PK data stored in the data store.
II. Definitions
(27) The disclosure, however, is not limited to these exemplary embodiments and applications or to the manner in which the exemplary embodiments and applications operate or are described herein. Moreover, the figures may show simplified or partial views, and the dimensions of elements in the figures may be exaggerated or otherwise not in proportion. In addition, as the terms "on," "attached to," "connected to," "coupled to," or similar words are used herein, one element (e.g., a material, a layer, a substrate, etc.) can be "on," "attached to," "connected to," or "coupled to" another element regardless of whether the one element is directly on, attached to, connected to, or coupled to the other element or there are one or more intervening elements between the one element and the other element. In addition, where reference is made to a list of elements (e.g., elements a, b, c), such reference is intended to include any one of the listed elements by itself, any combination of less than all of the listed elements, and/or a combination of all of the listed elements. Section divisions in the specification are for ease of review only and do not limit any combination of elements discussed.
(28) Unless otherwise defined, scientific and technical terms used in connection with the present teachings described herein shall have the meanings that are commonly understood by those of ordinary skill in the art. Further, unless otherwise required by context, singular terms shall include pluralities and plural terms shall include the singular. Generally, nomenclatures utilized in connection with, and techniques of, chemistry, biochemistry, molecular biology, pharmacology and toxicology are described herein are those well-known and commonly used in the art.
(29) As used herein, "substantially" means sufficient to work for the intended purpose. The term "substantially" thus allows for minor, insignificant variations from an absolute or perfect state, dimension, measurement, result, or the like such as would be expected by a person of ordinary skill in the field but that do not appreciably affect overall performance. When used with respect to numerical values or parameters or characteristics that can be expressed as numerical values, "substantially" means within ten percent.
(30) The term "ones" means more than one.
(31) As used herein, the term "plurality" can be 2, 3, 4, 5, 6, 7, 8, 9, 10, or more.
(32) As used herein, the phrase "Area Under the Curve" (AUC) may refer to an area of a curve that describes the variation of a therapeutic (drug) concentration in subject blood plasma as a function

of time post administration.
(33) As used herein, the phrase “Maximum Concentration” (C.sub.max) may refer to a maximum (or peak) serum concentration that a therapeutic (drug) achieves in a specified compartment or test area of the body after the drug has been administrated and before the administration of a second dose.
(34) As used herein, the phrase “Minimum Concentration” (C.sub.min) or “Trough Concentration” (C.sub.through) may refer to a minimum (or peak) serum concentration that a therapeutic (drug) achieves in a specified compartment or test area of the body after the drug has been administrated and before the administration of a second dose.
(35) As used herein, the term “Time of Maximum Concentration” (T.sub.max) may refer to a time when a maximum concentration is reached after a therapeutic (drug) dose is administered and before the next dose is administered.
(36) As used herein, the term “Half-life” (t.sub.1/2) may refer to a time that it takes for a concentration of a therapeutic (drug) in blood plasma to reach one-half of its steady-state value.
(37) As used herein, the term “mean residence time” (MRT) may refer to an average time that a therapeutic (drug) stays at the site of action.
(38) As used herein, the phrase “compartmental model” or “compartment modelling” may refer to mathematical modelling techniques used for predicting PK parameters (indicative of the ADME) of synthetic or natural therapeutic (drug) in a test subject by modelling concentrations of the therapeutic in different areas of the body. Within these mathematical models, the different areas of the body can be divided into parts, called compartments, where the therapeutic can be assumed to behave in the same manner.
(39) As used herein, the phrase “non-compartmental model” (NCA) may refer to model-independent techniques (meaning they do not rely upon assumptions about body compartments) used for predicting PK parameters (indicative of the ADME) of a therapeutic (drug) administered to a test subject. NCA enables the computation of PK parameters of a therapeutic (drug) from the time course of measured drug concentrations.
(40) As used herein, a “biologic” or “large molecule therapeutic” may refer to proteins and other biological macromolecules that have a therapeutic effect.
(41) As used herein, “therapeutics compound” or “small molecule therapeutic” may refer to any organic compound that affects a biologic process with a relatively low molecular weight, below 900 Daltons.
(42) As used herein, an “artificial neural network” or “neural network” (NN) may refer to mathematical algorithms or computational models. Neural networks can employ one or more layers of nonlinear units to predict an output for a received input. Some neural networks include one or more hidden layers in addition to an output layer. The output of each hidden layer is used as input to the next layer in the network, i.e., the next hidden layer or the output layer. Each layer of the network generates an output from a received input in accordance with current values of a respective set of parameters.
(43) A neural network may process information in two ways; when it is being trained it may be in “learning mode” and when it puts what it has learned into practice it may be in “inference (or prediction) mode.” Neural networks may learn through a feedback process called backpropagation which allows the network to adjust the weight factors (modifying its behavior) of the individual nodes in the intermediate hidden layers so that the output matches the outputs of the training data. In other words, a neural network may receive training data (learning examples) and automatically learn how to reach the correct output, even when it is presented with a new range or set of inputs. Examples of the types of neural networks, include, but are not limited to: Feedforward Neural Network (FNN), Recurrent Neural Network (RNN), Modular Neural Network (MNN), Convolutional Neural Network (CNN), Residual Neural Network (ResNet), etc.

III. Conventional Methods For Predicting PK Parameters

(44) FIG. **1** is a visualization graph that depicts a standard time-series concentration plot **100** and illustrates examples of typical PK parameters. As shown herein, the various types of PK parameters (e.g., C.sub.max **102**, t.sub.1/2 **104**, C.sub.min **106**, AUC **108**, etc.) are depicted on time-series concentration plot **100** for a drug after the drug has been administered to a test subject. C.sub.max **102** is a value that is indicative of the highest concentration of the drug in the test subject's blood. It is often of importance to know the maximum concentration, C.sub.max **102**, of a drug in the blood and the time it takes to reach this maximum, T.sub.max. For an intravenous bolus dose, the maximum concentration is obtained just after the drug is injected into the bloodstream, that is T.sub.max=0. In the case of extravascular (oral) administration, the concentration will not peak until after a while because of the absorption step. In general, the time it takes to reach the maximum can be found by differentiating the expression for C(t) for the system, with respect to t, set the derivative equal to zero and finally solve for T.sub.max.

(45) In plot **100**, t.sub.1/2 **104** or half-life represents the biological half-life of the drug in the test subject after administration. That is, the half-life of a drug is the time at which the drug has lost half its maximum concentration. C.sub.min **106** is the lowest concentration of the drug in the blood post administration. This PK parameter is oftentimes used in bioavailability and bioequivalence studies.

(46) AUC **108** or "Area Under the Curve" is a value that is indicative of the test subject's overall therapeutics exposure. AUC **108** can be evaluated using most types of models (compartmental or non-compartmental) and can be determined graphically based of a time-series of concentration measurements.

(47) FIG. **2** is an illustration of a time-series concentration plot **200**. As shown herein, time-series concentration plot **200** includes time-series concentration data **202** that is measured from blood samples drawn from a test subject at various time intervals that occur after the administration of a drug. The AUC **204** for time-series concentration plot **200** is comprised of a series can be estimated using the conventional "Trapezoidal Rule" whereby the AUC **204** under the times series concentration plot **200** is mathematically subdivided into a series of AUC segments **208**, which are trapezoids, that are bracketed by the time intervals at which the drug concentration is measured. The AUC **204** is computed as a sum-of-trapezoids calculation.

(48) FIG. **3** is an illustration of a workflow **300** for the sum-of-trapezoids calculation performed for the time-series concentration plot **200** in FIG. **2**. Data **302** includes the time points at which concentration measurements were generated. Plot **304** represents a trapezoid-based representation of time-series concentration plot **200** in FIG. **2** in which AUC **204** is divided into a series of trapezoids. Formula **306** identifies one manner in which the sum-of-trapezoids calculation can be performed to provide an estimation of the AUC. Thus, this AUC estimation is based on an approximation of the entire area by means of summing up individual subareas. This method of numerical integration generally assumes that the trajectory is partitioned until the last observation t.sub.n in n-1 sections.

(49) While this conventional approach is simple and can be fast, its accuracy relies on crucial presumptions that all the relevant measurements are available and that the time intervals between the measurements are appropriately short and condensed. Thus, as previously described, this type of approach has limitations with respect to its accuracy.

IV. Estimation of PK Parameters Using Deep Learning Neural Networks

(50) In contrast to the accuracy-impaired methods described in Section III, the various embodiments of the present disclosure provide methods and systems for accurately predicting one or more pharmacokinetic parameter values for an agent that is administered to a subject. These methods and systems can be implemented via computer software, hardware, firmware, or a combination thereof.

(51) FIG. **4** is a block diagram of a pharmacokinetic (PK) evaluation system **400** in accordance with one or more example embodiments. PK evaluation system **400** may be used to evaluate PK

effects resulting from the administration of a drug (e.g., therapeutic). In various embodiments, PK evaluation system **400** is trained based on observed data and then used to predict PK effects over time (including time beyond that for which observed data is provided or available). As previously described, a PK effect is a dose effect or an effect of the body on the drug in the body (e.g., drug concentration).

(52) PK evaluation system **400** may be used in various settings including, but not limited to, a research setting, clinical trial setting, a drug development setting, a hospital setting, or in some other type of setting. PK evaluation system **400** may receive and process input data **401** to generate a report **402** that describes and/or contains information based on these PK. Input data **401** may include, for example, lab measurements taken from the plasma of subjects, measurements from continuous monitoring devices (e.g., in hospital or home settings), or some other type of data. The report **402** may include, for example, a PK time course that is predicted for a certain number of days or weeks into the future based on a current dosing regimen. In some cases, report **402** may also include at least one of a corresponding PK time course that is predicted based on certain dosing interruptions and/or adjustments to the current dosing regimen. Report **402** may be generated for an individual subject in question or for the entire cohort of subjects in a clinical trial. In one or more examples, report **402** may include one or more recommended actions based on a predicted PK time course.

(53) PK evaluation system **400** includes computing platform **403**, data storage **404**, and display system **406**. Computing platform **403** may take various forms. In one or more embodiments, computing platform **403** includes a single computer (or computer system) or multiple computers in communication with each other. In other examples, computing platform **403** takes the form of a cloud computing platform. In some embodiments, computing platform **403** is referred to as a computing device/analytics server. In some embodiments, computing platform **403** can be a workstation, mainframe computer, distributed computing node (part of a “cloud computing” or distributed networking system), personal computer, mobile device, etc.

(54) Data storage **404** and display system **406** are each in communication with computing platform **403**. Data storage **404** may communicatively connected to computing platform **403** in various ways such as, but not limited to, a network connection that can be either a “hardwired” physical network connection (e.g., Internet, LAN, WAN, VPN, etc.) or a wireless network connection (e.g., Wi-Fi, WLAN, etc.).

(55) In some examples, data storage **404**, display system **406**, or both may be considered part of or otherwise integrated with computing platform **403**. Thus, in some examples, computing platform **403**, data storage **404**, and display system **406** may be separate components in communication with each other, but in other examples, some combination of these components may be integrated together.

(56) PK evaluation system **400** includes data manager **408** and prediction system **410** implemented in the computing platform **403**. Each of the data manager **408** and the neural network system **410** is implemented using hardware, software, firmware, or a combination thereof. Data manager **408** may also be referred to as a pharmacokinetic (PK) data simulation engine. Prediction system **410** may also be referred to as a pharmacokinetic (PK) prediction engine.

(57) It should be appreciated that the various components or engines of PK evaluation system **400** can be combined or collapsed into a single engine, component, or module, depending on the requirements of the particular application or system architecture. Moreover, in various embodiments, data storage **404**, data manager **408**, prediction system **410** or a combination thereof may comprise additional engines or components as needed by the particular application or system architecture.

(58) In various embodiments, the data manager **408** provides input data **401** to the neural network system **410**. This input data **401** may be retrieved from the data storage **404**, received from some other source, or a combination thereof. The input data **401** may have been generated via one or

more samples from one or more subjects. For example, a sample analyzer **420** can be communicatively connected to data storage **404** via one or more communications links (e.g., via a serial bus, a wireless network connection, etc.). Sample analyzer **420** can be configured to analyze blood samples from a test subject, such as test subject **422**, and determine the concentration of an agent (drug or biologic) in the blood. Time-series concentration data can be generated as blood samples are drawn from the subject **422** at set time intervals and analyzed using the sample analyzer **420**. In various embodiments, the time-series concentration data can then be stored in the data storage **404** for subsequent processing. In some embodiments, the time-series concentration data is fed as input data **401** to data manager **408** in real-time or near real-time.
(59) Prediction system **410** is a machine learning or deep learning prediction system. In one or more embodiments, prediction system **410** includes a neural network system **411**. The neural network system **411** includes one or more neural network models.
(60) When the neural network system **410** is being trained, the input data **401** takes the form of training data **416**. After training, the neural network system **410** may be used in practice to predict PK parameters and in these examples, the input data **401** may be referred to as testing data **418**. Thus, the type of input data **401** provided to the neural network system **410** may take different forms depending on whether the neural network system **410** is in a training mode or in a prediction mode.
(61) For example, the neural network system **410** may include a training module **412** and a prediction module **414**. The training module **412** may be used when the neural network system **410** is being trained (e.g., in a training mode). As one example, the training module **412** trains the neural network system **410** using training data **416** received from the data manager **408**. The prediction module **414** uses the neural network system **410** for prediction (e.g., in a prediction mode). For example, after the neural network system **410** has been trained, the prediction module **414** may be used in practice to predict PK parameters **424** using testing data **418** received from the data manager **408**.
(62) In some embodiments, data manager **408** simulates training data **416** used by prediction system **410**. For example, data manager **408** may be configured to use one or more compartmental-based pharmacokinetic models to generate training data **416** (e.g., a simulated training data collection). Training data **416** (the simulated training data collection) can be comprised of a plurality of discrete simulated time-series concentration datasets of one or more modeled agents (drug or biologic) and simulated values for one or more PK parameters that correspond to the plurality of discrete simulated time-series concentration datasets. Examples of the types of simulated pharmacokinetic parameters include, but are not limited to: AUC, C.sub.max, C.sub.min, C.sub.trough, T.sub.max, MRT, T.sub.last, or t.sub.1/2.
(63) In various embodiments, the compartmental-based pharmacokinetic models can be custom configured thru the input of one or more compartmental model parameters prior to the running of the model simulation to generate the simulated time-series concentration data. Examples of compartmental model parameters include, but are not limited to: dosing scheme (e.g., subcutaneous, oral, intravenous, etc.), molecular class (i.e., small molecule or large molecule), molecular species (e.g., drug, mAbs, protein, enzyme, etc.), compartmental model specific parameters (e.g., volume of distribution for the central compartment, absorption constants, elimination constants, etc.), modeled species (e.g., mammalian, rodents, human, non-human primates, etc.), modeled species demographics (e.g., age, weight, gender etc.), baseline albumin, baseline tumor size, etc.
(64) In various embodiments, the compartmental-based pharmacokinetic model can be a one-compartment pharmacokinetic model. For one-compartment pharmacokinetic models, the compartmental model parameters that can be set include, but are not limited to: volume of distribution for the central compartment, absorption rate into central compartment, elimination rate from the central compartment, etc.

(65) In various embodiments, the compartmental-based pharmacokinetic model can be a two-compartment model. For two-compartment models, the compartmental model parameters that can be set include, but are not limited to: volume of distribution for the central compartment, volume of distribution for the peripheral compartment, absorption rate into central compartment, elimination rate from the central compartment, inter-compartmental clearance between central and peripheral compartments, etc.
(66) In various embodiments, the compartmental-based pharmacokinetic model can be a Michaelis-Menten two-compartment model. For Michaelis-Menten PK two-compartment models, the compartmental model parameters that can be set include, but are not limited to: volume of distribution for the central compartment, volume of distribution for the peripheral compartment, absorption rate into central compartment, elimination rate from the central compartment, inter-compartmental clearance between central and peripheral compartments, Michaelis-Menton constants associated with the non-linear clearance mechanism, etc.
(67) In various embodiments, training data **416** is sent in real-time or near real-time to prediction system **410** for use in training neural network system **411**. In various embodiments, training data **416** is generated and stored in data storage **404** and then subsequently sent/retrieved to train neural network system **411**.
(68) Prediction system **410**, in some embodiments, can be configured to initialize a self-organizing neural network, to train the neural network using training data **416** and predict one or more PK parameters **424** based on the time-series concentration dataset of the agent obtained from the subject (and stored in data storage unit **706**). Examples of the types of predicted OK parameters include, but are not limited to: AUC, C.sub.max, C.sub.min, C.sub.trough, T.sub.max, MRT, T.sub.last, or t.sub.1/2.
(69) After the values for PK parameters **424** have been predicted, PK parameters **424** can be displayed as report **402** (e.g., a result or summary) on display system **406**. Display system **406** in some embodiments may be implemented at a client terminal **426** that is communicatively connected to computing platform **403**. In various embodiments, client terminal **426** can be a thin client computing device. In various embodiments, client terminal **426** can be a personal computing device having a web browser (e.g., INTERNET EXPLORER™, FIREFOX™, SAFARI™, etc.) that can be used to control the operation of data storage **404**, data manager **408**, prediction system **410**, or a combination thereof.
(70) FIG. **5** is a depiction of an exemplary NN system **500** and how it can be used to process time-series concentration data to predict PK parameters, in accordance with various embodiments. The NN system **500** shown in FIG. **5** can perform a ML-based methodology for PK parameter estimation or prediction whereby a NN model **510** is trained by a plurality of simulated PK training datasets (i.e., simulated time-series concentration data and their corresponding PK parameters) generated by PK compartmental model simulations and NCA modelling so as to enable the automation of NCA PK parameter modelling tasks by the trained NN model **510**. In various embodiments, the NN model **510** is a Feedforward Neural Network. In various embodiments, the NN model **510** is a Recurrent Neural Network. In various embodiments, the NN model **510** is a Modular Neural Network. It should be understood, however, that essentially any type of NN model **510** topology can be employed as long as it is capable of being trained by simulated PK training datasets to output accurate estimates or predictions of PK parameters **508** when fed real-life (not simulated) or real-time time-series concentration data **502**.
(71) In various embodiments, the NN model **510** is a Deep Learning NN model which is a class of machine learning algorithms that uses multiple (hidden) layers to progressively extract higher level features from the raw input. In various embodiments, the DL NN model is a Convolutional Neural Network. In various embodiments, the DL NN model is a Residual Neural Network.
(72) As shown in FIG. **5**, a NN model **510** can be comprised of an input layer **504**, one or more hidden layers **506** and an output layer **508**. In various embodiments, the input layer **504** is

configured to ingest times series concentration data **502** so that the data can be processed by the NN model **510**. In various embodiments, the input layer **504** can be configured to ingest additional inputs besides time-series concentration data, such as, but not limited to: the therapeutic (agent) being modeled, subject demographic information, disease status, etc.

(73) In various embodiments, the hidden layers **506** can be comprised of 2 or more, 5 or more, 4 or more, 5 or more, or greater than 5 layers of nodes. In various embodiments, the number of hidden layers **506** and/or the number of nodes in each hidden layer **506** (in the NN **510** model) is preset prior to training. In various embodiments, the number of hidden layers **506** and/or the number of nodes in each hidden layer **506** (in the NN **510** model) is automatically set by the NN model **510** when it is being trained.

(74) In various embodiments, the output layer **508** can be configured to output one or more types of PK parameters including, but not limited to: AUC, C.sub.max, C.sub.min, C.sub.trough, T.sub.max, MRT, T.sub.last, or t.sub.1/2.

(75) At a high level, NN model **510** operates by finding the correct mathematical manipulation (through "learning" using training data) to turn the input, either the input layer of input data or any a preceding layer, into the output of a subsequent layer, either the output layer of the output of the NN model **510** or the output of any given layer which will serve as the preceding layer to another layer. This occurs regardless of whether the mathematical manipulation is a linear relationship or a non-linear relationship. The NN model **510** "learns" by adjusting the weights (and optional thresholds) of the nodes in the various layers of the NN model to improve the accuracy of the result. Learning is complete when examining additional observations (of training data) does not usefully reduce the error rate or reduces the error rate to some pre-defined value. Complex NN models have many layers, hence the name "deep learning" neural networks.

(76) After learning (training) and during model use (inference, prediction), the NN model **510** serves to relate an input layer l input set of data to an output layer l output set of data via a mathematical function and the previously trained weights. During inference, the weights are fixed and the input data and numerical values of the weights together with the architecture of the layers of the NN model **510** (as introduced above), represent the mathematical function of how the input data and weights are computed together to calculate the outputs of the NN model **510**.

(77) FIG. **6** illustrates an exemplary high-level workflow of how a NN model can be trained during a learning phase and then puts what it has learned into practice to predict PK parameters, in accordance with various embodiments. As discussed above, neural network processes information in two ways; when it is being trained it is in a learning phase **602** and when it puts what it has learned into practice it is in an inference phase **604**.

(78) As shown herein, during the learning phase **602**, training data **606** is first preprocessed **608** and then it is iteratively fed into the NN model **610** through a backpropagation (which is one method of training) process that adjusts the weights (and optional thresholds) of the nodes in each of the various layers of the NN model **610** such that it converges to a predetermined (or preset) validation set error value **612** and/or test set error value **614**.

(79) In various embodiments, the preprocessing **608** of the training data **606** is a normalization of the time-series concentration data and the correlated NCA PK parameters that make up the training data **606**. In various embodiments, the preprocessing **608** of the training data. **606** filters out poor-quality datasets from the training data **606**.

(80) After the backpropagation process cycles through enough iterations such that the validation set error value **612** and the test set error value **614** converges to predetermined (preset) values, the fully trained NN model **618** can be used in the inference phase **604**. In the inference phase **604**, the clinically obtained time-series concentration data **616** (i.e., real-world clinical data) goes through a pre-processing step **620** and then this preprocessed data is fed to the trained NN model **618**, which processes the preprocessed data and outputs one or more predicted. PK parameter values **622**. The outputs can be PK parameters such as, but not limited to: AUC, C.sub.max, C.sub.min,

C.sub.trough, T.sub.max, MRT, T.sub.last, or t.sub.1/2.
(81) In various embodiments, the preprocessing **608** of the training data **606** is a normalization of the time-series concentration data and the correlated NCA PK parameters that make up the training data **606**. In various embodiments, the preprocessing **608** of the training data **606** filters out poor-quality datasets from the training data **606**.
(82) FIG. **7** is an illustration of a plot **700** showing percent validation error in accordance with an example embodiment. Plot **700** shows how the percent validation error, which is calculated as the predicted value versus actual value after running a validation dataset through NN model **610** during the training phase **602**, for predicted AUC.sub.inf and C.sub.max values decreases as the number of training iterations increases. It should be noted, however, that the improvements to the accuracy of the predictions eventually levels off as a function of the number of training iterations which suggests that that there is an optimal number of training iterations for NN model **610** and that there is a diminishing return in prediction accuracy when its surpassed.
(83) During the backpropagation process, the “forward pass” refers to the calculation process whereby the values of the output layer is calculated from the input data. The input data traverses through all nodes from the first to last layer. A loss function is calculated from the output values and then “backward pass” refers to the process of counting changes in weights (de facto learning), using a gradient descent algorithm (or similar). Computation is made from last layer, backward to the first layer. A backward and forward pass cycle together makes one iteration.
(84) FIG. **8** is an illustration of a plot **800** showing an example of training and validation loss as a function of the number of training iterations in accordance with an example embodiment. Plot **800** shows training and validation loss with respect to the initial NN model **610** and fully trained NN model **618** in FIG. **6**. Training loss is the error on the training set of data. Validation loss is the error after running the validation set of data through the trained network. Generally, the lower the loss, the better a model (unless the model has over-fitted to the training data). The loss is an interpretation as to how well the NN model **610** is doing for these two sets. Unlike accuracy, loss is not a percentage. It is a summation of the errors made for each example in training or validation sets.
(85) FIG. **9** is an illustration of an example training workflow **900** in accordance with various embodiments. Training workflow **900** is a high-level illustration of how simulated PK data can be generated and used to create training datasets.
(86) As depicted herein, simulated time-series concentration data **902** is generated using one or more types of PK compartmental models. This involves first setting (by inputting) one or more PK compartmental model simulation parameters and then running the PK compartmental model simulation to generate simulated time-series concentration data **902** for one or more existing or potential future therapeutic agents.
(87) Examples of PK compartmental model simulation parameters include, but are not limited to: dosing scheme (e.g., subcutaneous, oral, intravenous, etc.), molecular class (e.g., small molecule or large molecule), molecular species (e.g., drug, mAbs, protein, enzyme, etc.), compartmental model specific parameters (e.g., volume of distribution for the central compartment, absorption constants, elimination constants, etc.), modeled species (e.g., mammalian, rodents, human, non-human primates, etc.), modeled species demographics (e.g., age, weight, gender etc.), baseline albumin, baseline tumor size, etc.
(88) In various embodiments, a one-compartment model can be used to simulate the time-series concentration data **902**. For PK one-compartment models, the compartmental parameters that can be set include, but are not limited to: volume of distribution for the central compartment, absorption rate into central compartment, elimination rate from the central compartment, etc.
(89) In various embodiments, a two-compartment model can be used to simulate the time-series concentration data **902**. For PK two-compartment models, the compartmental parameters that can be set include, but are not limited to: volume of distribution for the central compartment, volume of

distribution for the peripheral compartment, absorption rate into central compartment, elimination rate from the central compartment, inter-compartmental clearance between central and peripheral compartments, etc.

(90) In various embodiments, a Michaelis-Menten PK two-compartment model can be used to simulate the time-series concentration data **902**. For Michaelis-Menten PK two-compartment models, the compartmental parameters that can be set include, but are not limited to: volume of distribution for the central compartment, volume of distribution for the peripheral compartment, absorption rate into central compartment, elimination rate from the central compartment, inter-compartmental clearance between central and peripheral compartments, Michaelis-Menton constants associated with the non-linear clearance mechanism, etc.

(91) After the simulated time-series concentration data **902** is generated, it is sparsely sampled **906** to form a subset of simulated time-series concentration training datasets **908** and each of the simulated time-series concentration training datasets **908** is further processed using non-compartmental analysis (NCA) **904** to calculate their corresponding simulated PK parameter values **910**. Examples of the types of simulated PK parameter values include, but are not limited to: AUC, C.sub.max, C.sub.min, C.sub.trough, T.sub.max, MRT, T.sub.last, or t.sub.1/2.

(92) FIG. **10** is an illustration of a series of plots **1000** for various forms of compartment modeling in accordance with various embodiments. Series of plots **1000** illustrate how compartmental modeling can be used to simulate time-series concentration data that is coupled with PK parameters predicted using NCA to form training datasets, in accordance with various embodiments.

(93) In the example illustrated in FIG. **10**, three different compartmental models (i.e., 1-Compartment **1010**, 2-Compartment **1012** and 2-Compartment Michaelis-Menten **1014**) were used to generate simulated time-series concentration datasets **1002**. The same compartmental model simulations were run using different therapeutic dosing scheme settings (i.e., bolus **1006** and oral/subcutaneous **1008**) to illustrate the effects of dosing scheme on the simulated time-series concentration datasets **1002**.

(94) As discussed above, after the simulated time-series concentration datasets **1002** are generated, they are further processed using NCA to calculate their corresponding simulated PK parameter values **1004**. In various embodiments, the simulated time-series concentration datasets are of one or more existing or potential future therapeutic agents.

V. Experimental Results

(95) The improved systems and methods, disclosed herein, were compared against conventional approaches to predicting a pharmacokinetic parameter value of an agent that is administered to a subject.

(96) FIG. **11** illustrates a set of plots **1100** that demonstrates the improved accuracy of estimating AUC and C.sub.max using the neural network approach described herein as compared to the conventional Trapezoidal Rule in accordance with an example embodiment.

(97) As shown herein, an AUC.sub.inf% Error histogram **1102** shows a plot of % Error (predicted vs actual) for AUC.sub.inf values determined using the NN methods disclosed herein and the conventional Trapezoidal Rule method. It is clear both visually and numerically (mean percentage error), that the AUC.sub.inf values determined using the NN methods were sizeably more accurate than the AUC.sub.inf values determined using the conventional Trapezoidal Rule method. For example, for AUC.sub.inf, the mean percentage error of the NN methods was about 9.1, while the Trapezoidal Rule mean percentage error was about 81.5.

(98) The observed significant improvement in accuracy of the NN methods over conventional methods was reinforced by the C.sub.max % Error histogram **1104** that plots % Error for C.sub.max values determined using the NN methods and the conventional general observation method. This is important because this shows that the improvement in accuracy using the NN predictive methods applies to determination of all PK parameters (e.g., C.sub.min, C.sub.trough, T.sub.max, or t.sub.1/2) and not just to the determination of AUC.sub.inf values. For C.sub.max,

the mean percentage error for the NN methods was about 2.2 the observed versus true mean percentage error was about 88.8.

VI. Example Methods

(99) FIG. **12** is an exemplary flowchart showing a method **1200** for predicting a PK parameter value of an agent (i.e., drug or biologic) administered to a subject (i.e., patient), in accordance with various embodiments.

(100) In step **1202**, a neural network is trained, by one or more processors, a neural network based on a simulated training data collection, the simulated training data collection comprising a simulated time-series concentration dataset and a simulated value for a pharmacokinetic parameter that corresponds to the simulated time-series concentration dataset. In some embodiments, step **1202** includes training the neural network based on multiple discrete simulated time-series concentration datasets, which may be, in turn, used to simulate values for one or more pharmacokinetic parameters for each simulated time-series concentration dataset of the multiple discrete simulated time-series concentration datasets.

(101) In one or more embodiments, the simulated value for the pharmacokinetic parameter corresponds to the entirety of the simulated time-series concentration dataset. In one or more embodiments, the simulated value for the pharmacokinetic parameter corresponds to a particular portion of the simulated time-series concentration dataset (e.g., one or more data points within the simulated time-series concentration dataset).

(102) In various embodiments, the neural network is a Feedforward Neural Network (FNN). In various embodiments, the neural network is a Recurrent Neural Network (RNN). In various embodiments, the neural network is a Modular Neural Network (MNN). In various embodiments, the neural network is a Convolutional Neural Network (CNN). In various embodiments, the neural network is a Residual Neural Network (ResNet).

(103) In various embodiments, the compartmental-based pharmacokinetic model is a one-compartment model. In various embodiments the compartmental-based pharmacokinetic model is a two-compartment model. In various embodiments, the compartmental-based pharmacokinetic model is a two-compartment Michaelis-Menten model.

(104) In step **1204**, a time-series concentration dataset of the agent obtained from a subject is received by the one or more processors. The agent may comprise, for example, a small molecule therapeutic or a biologic.

(105) In step **1206**, a value for the pharmacokinetic parameter is predicted, by the one or more processors, using the time-series concentration dataset and the neural network that has been trained. As discussed above, examples of the types of pharmacokinetic parameters for which values are predicted include, but are not limited to: AUC, C.sub.max, C.sub.min, C.sub.trough, T.sub.max, MRT, T.sub.last, or t.sub.1/2.

(106) FIG. **13** is an exemplary flowchart showing a method **1300** for training a neural network to predict a pharmacokinetic parameter value of an agent (i.e., drug or biologic), in accordance with various embodiments.

(107) In step **1302**, simulation parameters for a compartmental-based pharmacokinetic model are received by one or more processors. Examples of simulation parameters include, but are not limited to: dosing scheme (e.g., subcutaneous, oral, intravenous, etc.), molecular class (i.e., small molecule or large molecule), molecular species (e.g., drug, mAbs, protein, enzyme, etc.), compartmental model specific parameters (e.g., volume of distribution for the central compartment, absorption constants, elimination constants, etc.), modeled species (e.g., mammalian, rodents, human, non-human primates, etc.), modeled species demographics (e.g., age, weight, gender etc.), baseline albumin, baseline tumor size, etc.

(108) In various embodiments, the compartmental-based pharmacokinetic model is a one-compartment model. In various embodiments the compartmental-based pharmacokinetic model is a two-compartment model. In various embodiments, the compartmental-based pharmacokinetic

model is a two-compartment model with Michaelis-Menten clearance.

(109) In step **1304**, a simulated training data collection is generated, by the one or more processors, using the compartmental-based pharmacokinetic model. The simulated training data collection includes a simulated time-series concentration dataset (or a plurality of simulated time-series concentration datasets) and a simulated value for at least one pharmacokinetic parameter that corresponds to the simulated time-series concentration dataset (or the plurality of simulated time-series concentration datasets). In one or more embodiments, simulated values for the one or more pharmacokinetic parameters are generated for each simulated time-series concentration dataset of the plurality of simulated time-series concentration datasets.

(110) The compartmental-based pharmacokinetic model is encoded with the received simulation parameters. Examples of the types of pharmacokinetic parameters for which simulated values are generated include, but are not limited to: AUC, C.sub.max, C.sub.min, C.sub.trough, T.sub.max, MRT, T.sub.last, or t.sub.1/2.

(111) In step **1306**, a neural network is trained, by the one or more processors, with the simulated training data collection using the one or more processors. The trained neural network may then be used to predict a pharmacokinetic parameter value of an agent.

(112) In various embodiments, the neural network is a Feedforward Neural Network (FNN). In various embodiments, the neural network is a Recurrent Neural Network (RNN). In various embodiments, the neural network is a Modular Neural Network (MNN). In various embodiments, the neural network is a Convolutional Neural Network (CNN). In various embodiments, the neural network is a Residual Neural Network (ResNet).

(113) FIG. **14** is an exemplary flowchart showing a method **1400** for predicting a PK parameter value of an agent (e.g., drug, biologic, etc.) administered to a subject (i.e., patient), in accordance with various embodiments.

(114) In step **1402**, a time-series concentration dataset of an agent obtained from a subject is received by one or more processors.

(115) In step **1404**, a value for a pharmacokinetic parameter is predicted, by the one or more processors, based on the time-series concentration dataset and a neural network that has been trained using a simulated training data collection. The neural network may be trained via, for example, without limitation, method **1300** in FIG. **13**. The neural network may be trained using a simulated training data collection comprising a plurality of simulated time-series concentration datasets of one or more agents and simulated values for one or more pharmacokinetic parameters for each simulated time-series concentration dataset of the plurality of simulated time-series concentration datasets.

(116) In various embodiments, the neural network is a Feedforward Neural Network (FNN). In various embodiments, the neural network is a Recurrent Neural Network (RNN). In various embodiments, the neural network is a Modular Neural Network (MNN). In various embodiments, the neural network is a Convolutional Neural Network (CNN). In various embodiments, the neural network is a Residual Neural Network (ResNet).

VII. Computer-Implemented System

(117) FIG. **15** is a block diagram that illustrates a computer system **1500**, upon which embodiments of the present teachings may be implemented. In various embodiments of the present teachings, computer system **1500** can include a bus **1502** or other communication mechanism for communicating information, and a processor **1504** coupled with bus **1502** for processing information. In various embodiments, computer system **1500** can also include a memory, which can be a random access memory (RAM) **1506** or other dynamic storage device, coupled to bus **1502** for determining instructions to be executed by processor **1504**. Memory also can be used for storing temporary variables or other intermediate information during execution of instructions to be executed by processor **1504**. In various embodiments, computer system **1500** can further include a read only memory (ROM) **1508** or other static storage device coupled to bus **1502** for storing static

information and instructions for processor **1504**. A storage device **1510**, such as a magnetic disk or optical disk, can be provided and coupled to bus **1502** for storing information and instructions.

(118) In various embodiments, computer system **1500** can be coupled via bus **1502** to a display **1512**, such as a cathode ray tube (CRT) or liquid crystal display (LCD), for displaying information to a computer user. An input device **1514**, including alphanumeric and other keys, can be coupled to bus **1502** for communicating information and command selections to processor **1504**. Another type of user input device is a cursor control **1516**, such as a mouse, a trackball or cursor direction keys for communicating direction information and command selections to processor **1504** and for controlling cursor movement on display **1512**. This input device **1514** typically has two degrees of freedom in two axes, a first axis (i.e., x) and a second axis (i.e., y), that allows the device to specify positions in a plane. However, it should be understood that input devices **1514** allowing for 3 dimensional (x, y and z) cursor movement are also contemplated herein.

(119) Consistent with certain implementations of the present teachings, results can be provided by computer system **1500** in response to processor **1504** executing one or more sequences of one or more instructions contained in memory **1506**. Such instructions can be read into memory **1506** from another computer-readable medium or computer-readable storage medium, such as storage device **1510**. Execution of the sequences of instructions contained in memory **1506** can cause processor **1504** to perform the processes described herein. Alternatively hard-wired circuitry can be used in place of or in combination with software instructions to implement the present teachings. Thus implementations of the present teachings are not limited to any specific combination of hardware circuitry and software.

(120) The term “computer-readable medium” (e.g., data store, data storage, etc.) or “computer-readable storage medium” as used herein refers to any media that participates in providing instructions to processor **1504** for execution. Such a medium can take many forms, including but not limited to, non-volatile media, volatile media, and transmission media. Examples of non-volatile media can include, but are not limited to, optical, solid state, magnetic disks, such as storage device **1510**. Examples of volatile media can include, but are not limited to, dynamic memory, such as memory **1506**. Examples of transmission media can include, but are not limited to, coaxial cables, copper wire, and fiber optics, including the wires that comprise bus **1502**.

(121) Common forms of computer-readable media include, for example, a floppy disk, a flexible disk, hard disk, magnetic tape, or any other magnetic medium, a CD-ROM, any other optical medium, punch cards, paper tape, any other physical medium with patterns of holes, a RAM, PROM, and EPROM, a FLASH-EPROM, any other memory chip or cartridge, or any other tangible medium from which a computer can read.

(122) In addition to computer readable medium, instructions or data can be provided as signals on transmission media included in a communications apparatus or system to provide sequences of one or more instructions to processor **1504** of computer system **1500** for execution. For example, a communication apparatus may include a transceiver having signals indicative of instructions and data. The instructions and data are configured to cause one or more processors to implement the functions outlined in the disclosure herein. Representative examples of data communications transmission connections can include, but are not limited to, telephone modem connections, wide area networks (WAN), local area networks (LAN), infrared data connections, NFC connections, etc.

VIII. Exemplary Embodiments

(123) Various exemplary embodiments are described herein.

(124) In one or more embodiments, a method is provided for training a neural network to predict at least one pharmacokinetic parameter of an agent. One or more processors receive a simulation parameter for a compartmental-based pharmacokinetic model. The one or more processors generate a simulated training data collection using the compartmental-based pharmacokinetic model. The simulated training data collection comprises a simulated time-series concentration dataset of one or

more agents and a simulated value for a pharmacokinetic parameter that corresponds to the simulated time-series concentration dataset. The compartmental-based pharmacokinetic model is encoded with the received simulation parameters. The one or more processors then train a neural network with the simulated training data collection.

(125) In one or more embodiments, a system for training a neural network to predict a pharmacokinetic parameter value of an agent is disclosed. The system includes a data store, a computing device, and a display system. The data store stores simulation parameters for a compartmental-based pharmacokinetic model. The computing device is communicatively connected to the data store and hosts a pharmacokinetic data simulation engine and a pharmacokinetic prediction engine. The pharmacokinetic data simulation engine is configured to use the compartmental-based pharmacokinetic models to generate a simulated training data collection. The simulated training data collection comprises a simulated time-series concentration dataset of one or more agents and a simulated value for a pharmacokinetic parameter that corresponds to the simulated time-series concentration dataset. The pharmacokinetic prediction engine is configured to train a neural network using the simulated training data collection.

(126) In one or more embodiments, a system for predicting a pharmacokinetic parameter value of an agent administered to a subject comprises a data store for storing a time-series concentration dataset of an agent obtained from a subject; a computing device communicatively connected to the data store that comprises: a pharmacokinetic prediction engine configured to train a neural network and predict a pharmacokinetic parameter value based on the time-series concentration dataset of the agent obtained from the subject, wherein the neural network was trained using a simulated training data collection comprising a plurality of simulated time-series concentration datasets of one or more agents as well as corresponding simulated pharmacokinetic parameter values; and a display communicatively connected to the computing device and configured to display a report containing the predicted pharmacokinetic parameter value.

(127) In one or more embodiments, a non-transitory computer-readable medium storing computer instructions for predicting at least one pharmacokinetic parameter value of an agent administered to a subject comprising: training, by one or more processors, a neural network based on a simulated training data collection, the simulated training data collection comprising a simulated time-series concentration dataset and a simulated value for a pharmacokinetic parameter that corresponds to the simulated time-series concentration dataset; receiving, by the one or more processors, a time-series concentration dataset of the agent obtained from a subject; and predicting, by the one or more processors, a value for the pharmacokinetic parameter using the time-series concentration dataset and the neural network that has been trained.

(128) In one or more embodiments, a non-transitory computer-readable medium storing computer instructions for training neural network to predict at least one pharmacokinetic parameter of an agent, comprising: receiving a simulation parameter for a compartmental-based pharmacokinetic model; generating a simulated training data collection using the compartmental-based pharmacokinetic model, wherein the simulated training data collection comprises a simulated time-series concentration dataset of one or more agents and a simulated value for a pharmacokinetic parameter that corresponds to the simulated time-series concentration dataset; encoding the compartmental-based pharmacokinetic model with the received simulation parameters; and training a neural network with the simulated training data collection.

(129) In one or more embodiments, a non-transitory computer-readable medium storing computer instructions for predicting at least one pharmacokinetic parameter of an agent administered to a subject, comprising: receiving, by one or more processors, a time-series concentration dataset of the agent obtained from a subject; and predicting, by the one or more processors, a value for a pharmacokinetic parameter based on the time-series concentration dataset and a neural network that has been trained using a simulated training data collection.

(130) In one or more embodiments, the neural network of one or more of the exemplary

embodiments described herein is a convolutional neural network in some embodiments.
(131) In one or more embodiments, the simulation parameter of one or more of the exemplary embodiments described herein is a class of agent being simulated.
(132) In one or more embodiments, the agent of one or more of the exemplary embodiments described herein is one of a small molecule therapeutic or a biologic.
(133) In one or more embodiments, the simulation parameter of one or more of the exemplary embodiments described herein is a dosing scheme for the agent.
(134) In one or more embodiments, the dosing scheme of one or more of the exemplary embodiments described herein is one of an intravenous administration of the agent or an oral administration of the agent.
(135) In one or more embodiments, the compartmental-based pharmacokinetic model of one or more of the exemplary embodiments described herein comprises at least one of a one-compartment model, a two-compartment model, or a Michaelis-Menten model.
(136) In one or more embodiments, the data store and the computing device of one or more of the exemplary embodiments described herein are part of an integrated apparatus.
(137) In one or more embodiments, the data store of one or more of the exemplary embodiments described herein is hosted by a different device than the computing device of one or more of the exemplary embodiments described herein.
(138) In one or more embodiments, the data store and the computing device of one or more of the exemplary embodiments described herein are part of a distributed network system.
IX. Additional Considerations
(139) It should be appreciated that the methodologies described herein flow charts, diagrams and accompanying disclosure can be implemented using computer system **1300** as a standalone device or on a distributed network of shared computer processing resources such as a cloud computing network.
(140) The methodologies described herein may be implemented by various means depending upon the application. For example, these methodologies may be implemented in hardware, firmware, software, or any combination thereof. For a hardware implementation, the processing unit may be implemented within one or more application specific integrated circuits (ASICs), digital signal processors (DSPs), digital signal processing devices (DSPDs), programmable logic devices (PLDs), field programmable gate arrays (FPGAs), processors, controllers, micro-controllers, microprocessors, electronic devices, other electronic units designed to perform the functions described herein, or a combination thereof.
(141) In various embodiments, the methods of the present teachings may be implemented as firmware and/or a software program and applications written in conventional programming languages such as C, C++, Python, etc. If implemented as firmware and/or software, the embodiments described herein can be implemented on a non-transitory computer-readable medium in which a program is stored for causing a computer to perform the methods described above. It should be understood that the various engines described herein can be provided on a computer system, such as computer system **1100**, whereby processor **1104** would execute the analyses and determinations provided by these engines, subject to instructions provided by any one of, or a combination of, memory components **1106**/**1108**/**1110** and user input provided via input device **1114**.
(142) While the present teachings are described in conjunction with various embodiments, it is not intended that the present teachings be limited to such embodiments. On the contrary, the present teachings encompass various alternatives, modifications, and equivalents, as will be appreciated by those of skill in the art.
(143) In describing various embodiments, the specification may have presented a method and/or process as a particular sequence of steps. However, to the extent that the method or process does not rely on the particular order of steps set forth herein, the method or process should not be limited

to the particular sequence of steps described, and one skilled in the art can readily appreciate that the sequences may be varied and still remain within the spirit and scope of the various embodiments.

Claims

1. A system for predicting at least one pharmacokinetic parameter of an agent administered to a subject, comprising: one or more processors; and a non-transitory memory coupled to the processors comprising instructions executable by the processors, the processors operable when executing the instructions to: access a blood sample obtained from the subject; generate a time-series concentration data associated with the agent based on an analysis of the blood sample by a sample analyzer coupled to the system; execute a neural network on the time-series concentration data to predict a value for the at least one pharmacokinetic parameter, wherein predicting the value for the at least one pharmacokinetic parameter comprises predicting the value based on weights of a plurality of nodes across a plurality of layers associated with the neural network and the time-series concentration data, wherein the neural network has been trained using a simulated training data collection comprising a plurality of simulated time-series concentration data and a plurality of simulated values for the at least one pharmacokinetic parameter that correspond to the plurality of simulated time-series concentration data, respectively, and where the simulated training data collection is generated based on running simulations by one or more compartment-based pharmacokinetic models based on customized model parameters associated with the one or more compartment-based pharmacokinetic models; and display a report containing the value predicted for the at least one pharmacokinetic parameter.
2. The system of claim 1, wherein the one or more compartmental-based pharmacokinetic models comprise at least one of a one-compartment model, a two-compartment model, or a Michaelis-Menten model.
3. The system of claim 1, wherein the neural network comprises a convolutional neural network.
4. The system of claim 1, wherein the agent comprises a small molecule therapeutic or a biologic.
5. The system of claim 1, wherein the processors are further operable when executing the instructions to store the time-series concentration data in a data store, and wherein the data store and the system are part of an integrated apparatus.
6. The system of claim 1, wherein the processors are further operable when executing the instructions to store the time-series concentration data in a data store, and wherein the data store is hosted by a different device than the system.
7. A method for predicting at least one pharmacokinetic parameter of an agent administered to a subject, comprising: accessing a blood sample obtained from the subject; generating, by a sample analyzer coupled to one or more processors, a time-series concentration data associated with the agent based on an analysis of the blood sample; and executing, by the one or more processors, a neural network on the time-series concentration data to predict a value for the at least one pharmacokinetic parameter, wherein predicting the value for the at least one pharmacokinetic parameter comprises predicting the value based on weights of a plurality of nodes across a plurality of layers associated with the neural network and the time-series concentration data, wherein the neural network has been trained using a simulated training data collection comprising a plurality of simulated time-series concentration data and a plurality of simulated values for the at least one pharmacokinetic parameter that correspond to the plurality of simulated time-series concentration data, respectively, and where the simulated training data collection is generated based on running simulations by one or more compartment-based pharmacokinetic models based on customized model parameters associated with the one or more compartment-based pharmacokinetic models.
8. The method of claim 7, wherein the neural network is a convolutional neural network.
9. The method of claim 7, wherein the at least one pharmacokinetic parameter comprises at least

one of an area under the curve (AUC), a minimum concentration value (C.sub.min), or a maximum concentration value (C.sub.max).
10. The method of claim 7, wherein the agent comprises a small molecule therapeutic or a biologic.
11. The method of claim 7, wherein: training the neural network comprises feeding the simulated training data collection into the neural network through a backpropagation process that adjusts the weights of the plurality of nodes across the plurality of layers associated with the neural network; and generating the simulated training data collection comprises: setting the customized model parameters associated with the one or more compartment-based pharmacokinetic models, wherein the customized model parameters comprise at least a dosing scheme associated with the agent, a molecular species associated with the agent, and a species associated with the subject; running the simulations by the one or more compartment-based pharmacokinetic models based on the customized model parameters to generate the plurality of simulated time-series concentration data; and using non-compartment analysis (NCA) to generate the plurality of simulated values for the at least one pharmacokinetic parameter that correspond to the plurality of simulated time-series concentration data, respectively.